STAGE 1: Study Ingestion, Quality Control & Selection

Uniform Count Ingestion

- NCBI GEO-generated raw counts (one pipeline)
- GEO supplementary files, recount3, local files
- Any `SummarizedExperiment` via `mx_from_se()`

`mx_fetch_geo()` `mx_load_local()`

Data-Level QC (10 criteria)

- Each criterion: pass / fail / not evaluable
- Replicates, depth, detection, labels, raw counts
- Exclusion on failed required criteria

`mx_qc_study()` ⟶ `mx_filter_studies()`

STAGE 2: Identifier Harmonization & Per-Study Differential Expression

Gene Harmonization

- Symbol / Ensembl / Entrez
- Version stripping
- Duplicate collapsing

`mx_reannotate()`

Per-Study DE (raw counts)

- **DESeq2** / **edgeR**-QL / **limma-voom**
- Covariates, explicit contrasts
- log2FC, log2-scale SE, z

`mx_de_all()`

Pre-Pooling Checks

- Fold-change concordance
- Flipped-contrast detection
- Gene coverage per study

`mx_study_concordance()`

STAGE 3: Gene-Level Meta-Analysis (`mx_meta()`)

Random Effects

REML / PM / DL

- Optional HKSJ
- τ^2, I^2, 95% CI
- Prediction interval

Fixed Effect

Inverse-Variance

- Assumes $\tau^2 = 0$
- Weight $w_i = 1/s_i^2$
- Replicate studies

P-Value Pooling

Directional

- Fisher (Owen 2009)
- Stouffer, inv.-normal
- minP

Adaptive & Checks

AWmeta, rOP, maxP

- Study selection
- Leave-one-out
- Egger / Begg

STAGE 4: Pathways, Downstream Biology, Visualization & Reporting

Pathways & Biology

- fgsea GSEA / ORA on the meta-signature (`mx_pathway_meta()`)
- Per-study GSEA combined (`mx_pathway_studies()`)
- TF activity, cell composition, co-expression modules
- Drug-signature reversal (LINCS L1000)

Figures & Reports

- Volcano, forest (CI + PI), funnel, heterogeneity
- S4 classes built on `SummarizedExperiment`
- HTML report with methods text, CSV/Excel export

`mx_report()` `mx_export()`